

# ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

|         |                     |
|---------|---------------------|
| TxID    | 2483483             |
| ToLID   | <b>xgChaSegu1</b>   |
| Species | Charonia sequeenzae |
| Class   | Gastropoda          |
| Order   | Littorinimorpha     |

| Genome Traits     | Expected              | Observed      |
|-------------------|-----------------------|---------------|
| Haploid size (bp) | 1,016,308,473         | 4,994,661,906 |
| Haploid Number    | 17 (source: ancestor) | 35            |
| Ploidy            | 2 (source: ancestor)  | 2             |
| Sample Sex        | Unknown               | Unknown       |

## EBP metrics summary and curation notes

Obtained EBP quality metric for collapsed: 5.8.Q33

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

- . Observed Haploid size (bp) has >20% difference with Expected
- . Observed Haploid Number is different from Expected
- . QV value is less than 40 for collapsed
- . Kmer completeness value is less than 90 for collapsed
- . BUSCO single copy value is less than 90% for collapsed
- . BUSCO duplicated value is more than 5% for collapsed
- . More than 1000 gaps/Gbp for collapsed
- . Not 90% of assembly in chromosomes for collapsed

### Curator notes

- . Interventions/Gb: 187
- . Contamination notes: ""
- . Other observations: "The assembly of Charonia sequeenzae (xgChaSegu1) is based on 58.9X ONT data and 137.09X Arima Hi-C data generated as part of the European Reference Genome Atlas (ERGA, <https://www.erga-biodiversity.eu/>) via the Biodiversity Genomics Europe project (BGE, <https://biodiversitygenomics.eu/>). The assembly process included the following steps: initial ONT assembly generation with Hifiasm, removal of contaminant sequences using Context, removal of haplotypic duplications using purge\_dups, and Hi-C-based scaffolding with YaHS. In total, 888 contigs were identified as contaminants (bacterial, archaeal, or viral), totaling 57.417 Mb (with the largest being 0.479 Mb). Additionally, 17814 regions totaling 414.712 Mb (with the largest being 0.353 Mb) were identified as haplotypic duplications and removed.

Mitochondrial genome was assembled using ptGAUL. Finally, the primary assembly was analyzed and manually improved using Pretext. Chromosome-scale scaffolds confirmed by Hi-C data were named in order of size. "

# Quality metrics table

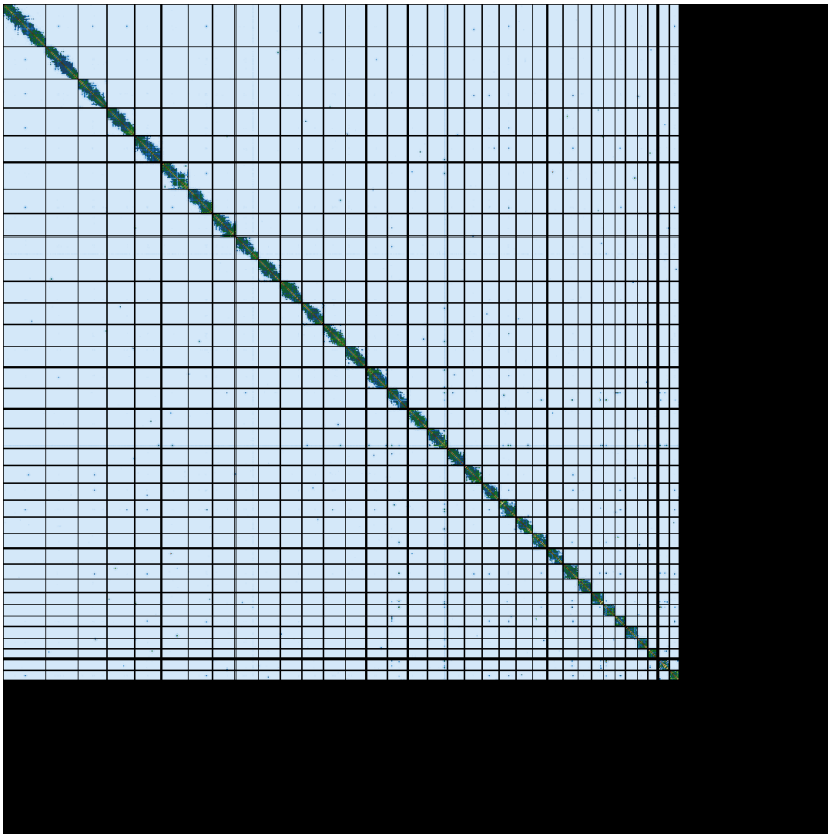
| Metrics      | Pre-curation collapsed | Curated collapsed |
|--------------|------------------------|-------------------|
| Total bp     | 5,002,866,701          | 4,994,661,906     |
| GC %         | 43.82                  | 43.82             |
| Gaps/Gbp     | 4,413.07               | 4,473.78          |
| Total gap bp | 2,207,800              | 2,294,600         |
| Scaffolds    | 17,889                 | 14,251            |
| Scaffold N50 | 79,325,356             | 112,930,338       |
| Scaffold L50 | 21                     | 18                |
| Scaffold L90 | 2,372                  | 2,028             |
| Contigs      | 39,967                 | 36,596            |
| Contig N50   | 256,834                | 257,473           |
| Contig L50   | 4,382                  | 4,367             |
| Contig L90   | 21,931                 | 21,801            |
| QV           | 33.432                 | 33.434            |
| Kmer compl.  | 86.7935                | 86.6938           |
| BUSCO sing.  | 70.2%                  | 68.8%             |
| BUSCO dupl.  | 20.5%                  | 19.3%             |
| BUSCO frag.  | 6.0%                   | 8.2%              |
| BUSCO miss.  | 3.3%                   | 3.7%              |

Warning! BUSCO versions or lineage datasets are not the same across results:

BUSCO: 6.0.0 (euk\_genome\_min, miniprot) / Lineage: mollusca\_odb12 (genomes:36, BUSCOs:4421)

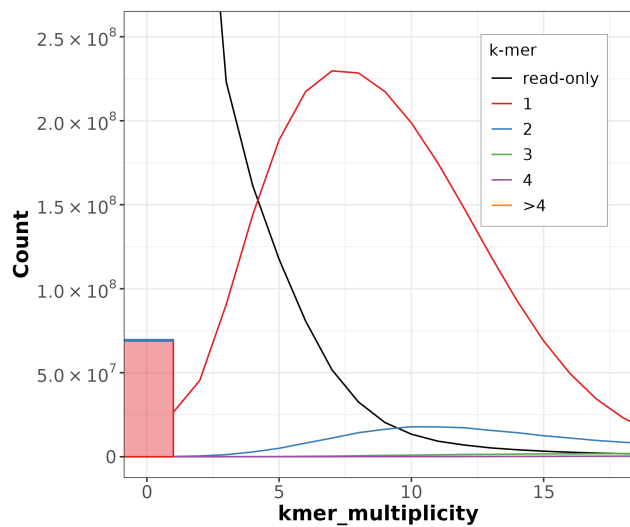
BUSCO: 5.8.2 (euk\_genome\_min, miniprot) / Lineage: mollusca\_odb12 (genomes:36, BUSCOs:4421)

# HiC contact map of curated assembly

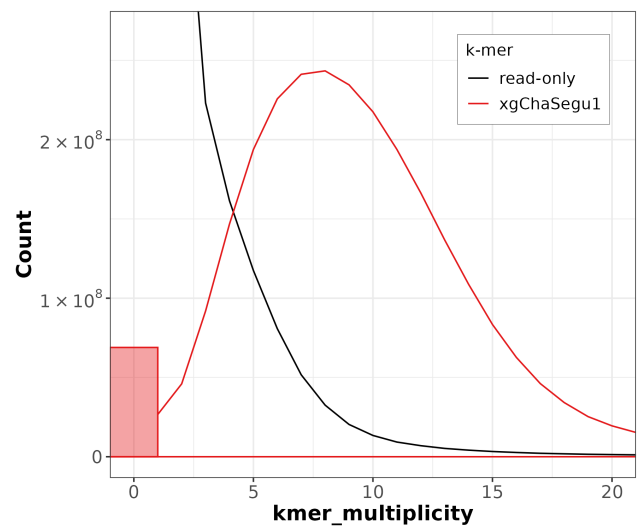


collapsed [\[LINK\]](#)

# K-mer spectra of curated assembly

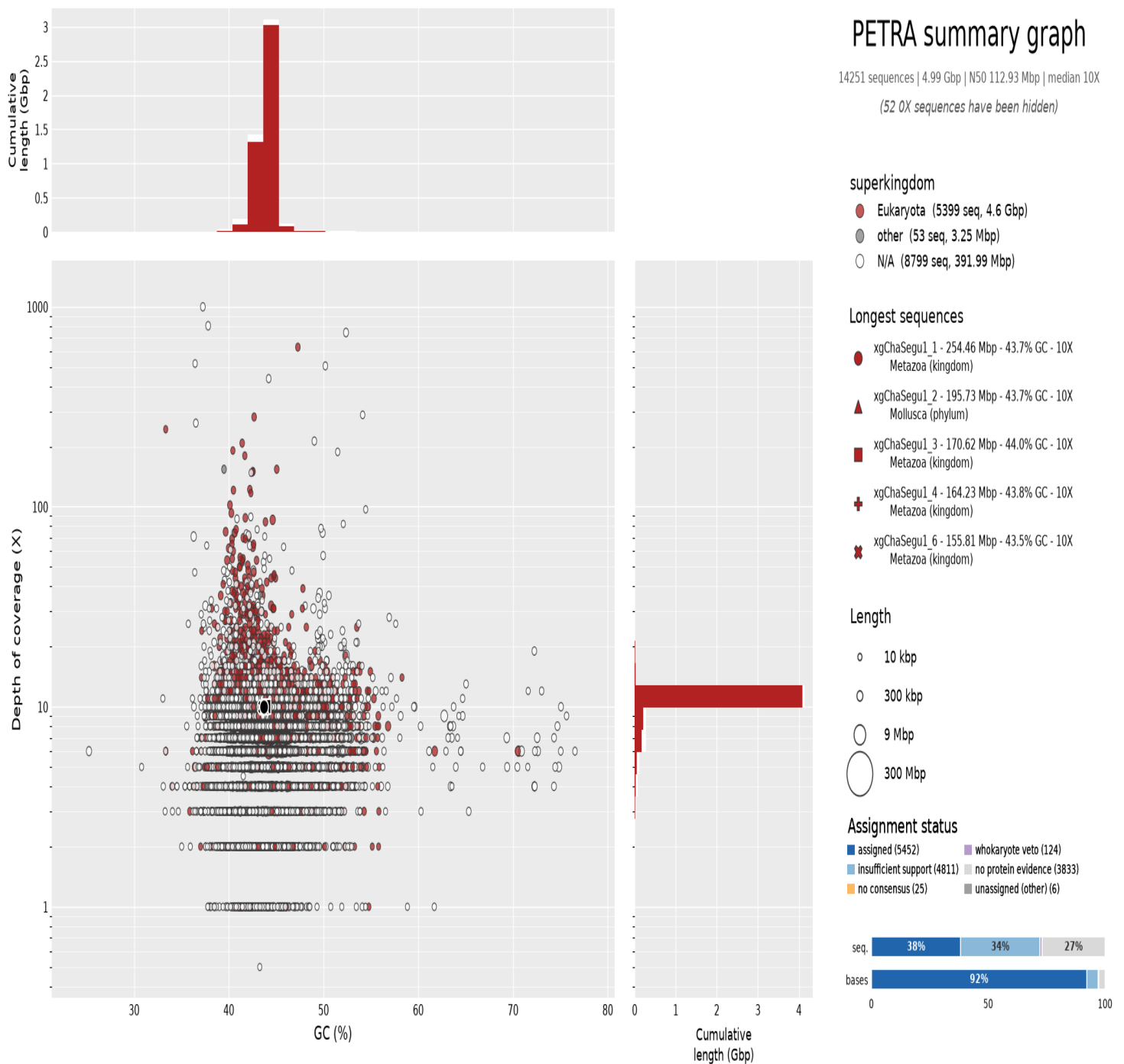


Distribution of k-mer counts per copy numbers found in asm



Distribution of k-mer counts coloured by their presence in reads/assemblies

# Post-curation contamination screening



**collapsed.** Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

# Data profile

| Data     | Long reads | Arima |
|----------|------------|-------|
| Coverage | 58         | 137   |

## Assembly pipeline

- **Hifiasm**
  - |\_ *ver*: 0.19.5-r593
  - |\_ *key param*: NA
- **purge\_dups**
  - |\_ *ver*: 1.2.5
  - |\_ *key param*: NA
- **YaHS**
  - |\_ *ver*: 1.2
  - |\_ *key param*: NA

## Curation pipeline

- **PretextMap**
  - |\_ *ver*: 0.2.4
  - |\_ *key param*: NA
- **PretextView**
  - |\_ *ver*: 1.0.7
  - |\_ *key param*: NA

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